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DATA STORYTELLING & STATISTICAL VALIDATION

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GOOD DATA TELLS A STORY. GREAT ANALYSIS TURNS THAT STORY INTO ACTION.

BUSINESS OBJECTIVE

What Are We Trying to Understand?

The objective of this analysis is to transform sales data into clear, actionable business insights.

Key Questions:

- Which product categories perform best?
- What customer patterns can we identify?
- Which factors are associated with Total Sales?
- Can statistical testing validate an important business finding?
- What actions can the business take based on the results?

DATA & ANALYTICAL APPROACH

From Raw Data to Insights

Dataset Overview

- ~1,000 sales records
- 18 columns
- 947 customers
- Customer, product, pricing and transaction information

Analytical Process

Data Cleaning → EDA → SQL Analysis → Power BI → Statistical Validation → Business Recommendations

Tools Used

Python | Pandas | SQL | Excel | Power BI

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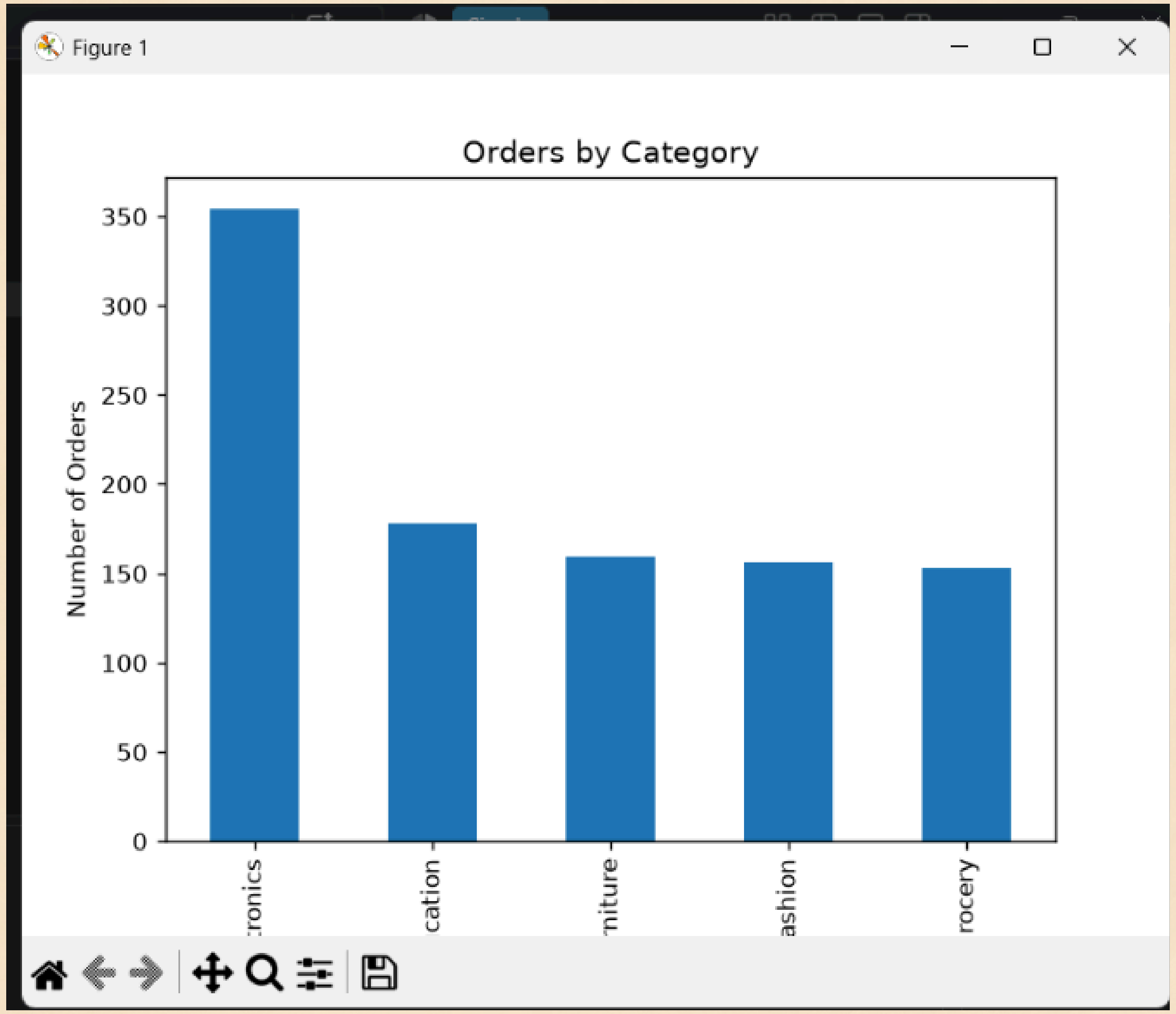


SALES PERFORMANCE

Electronics Leads Order Volume

Electronics recorded 354 orders, making it the strongest category by order volume.

Category Performance

- Electronics — 354 orders
- Education
- Furniture
- Fashion
- Grocery

Key Insight

Electronics stands out as the most active category and represents an important opportunity for the business to maintain and expand its performance.

UNDERSTANDING THE CUSTOMERS

Who Are the Customers?

Gender Distribution

- Male — 511
- Female — 489

Age Profile

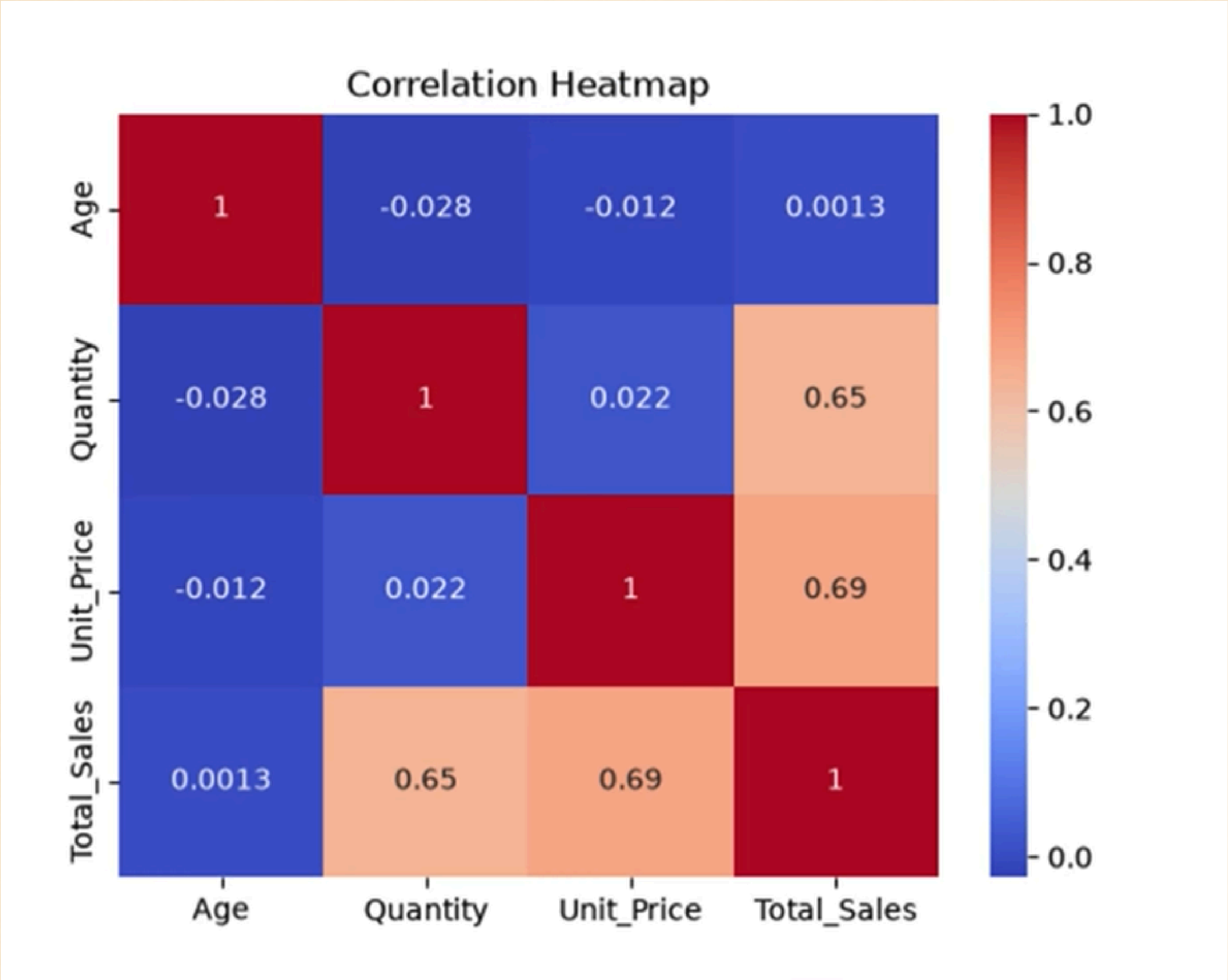
- Minimum age — 18
- Median age — 41
- Maximum age — 65

Key Insight

The customer base is relatively balanced by gender, while age alone does not show a strong relationship with spending.

Business Meaning

Customer targeting should consider purchase behavior and transaction patterns, not demographics alone.



WHAT DRIVES TOTAL SALES?

Quantity & Unit Price Show Stronger Relationships

Factor	**Correlation with Total Sales**
Quantity	0.65
Unit Price	0.69
Age	Weak relationship

Key Insight

Unit Price and Quantity show meaningful positive relationships with Total Sales.

Business Meaning

Higher purchase quantities and product pricing are important factors to consider when developing revenue strategies.

BUSINESS HYPOTHESIS

Can We Statistically Validate the Finding?

Business Question

Do Electronics orders have a significantly different average Total Sales compared with non-Electronics orders?

H_0 — Null Hypothesis

There is no significant difference in average Total Sales between Electronics and non-Electronics orders.

H_1 — Alternative Hypothesis

There is a significant difference in average Total Sales between Electronics and non-Electronics orders.

Statistical Test

Independent Two-Sample T-Test

Significance Level: $\alpha = 0.05$

STATISTICAL VALIDATION

Metric	Result
Electronics Average Sales	₹143,442.32
Non-Electronics Average Sales	₹137,183.99
T-statistic	0.818
P-value	0.414
95% Confidence Interval	₹−8,764.16 to ₹21,280.82

TATISTICAL DECISION

P-VALUE = 0.414 > 0.05

➡️ FAIL TO REJECT H_0

BUSINESS CONCLUSION
THERE IS NOT ENOUGH STATISTICAL EVIDENCE TO CONCLUDE THAT ELECTRONICS ORDERS HAVE A SIGNIFICANTLY DIFFERENT AVERAGE TOTAL SALES COMPARED WITH NON-ELECTRONICS ORDERS.

RIMBERIO TEAM

BUSINESS RECOMMENDATIONS

Turning Insights into Action

Strengthen High-Performing Categories
Continue monitoring Electronics and identify opportunities for growth.

Optimize Pricing
Use pricing analysis to understand how price changes may influence sales.

Encourage Higher Purchase Volume
Use bundles, promotions and volume-based offers to increase quantities purchased.

Improve Customer Targeting
Combine demographics with actual purchasing behavior for better segmentation.

Use Statistical Validation
Validate important business assumptions before making major decisions.

CONCLUSION

From Data to Decisions

The analysis transformed sales data into actionable business insights.

Key Takeaways:

- Electronics recorded the highest order volume.
- Customer distribution was relatively balanced by gender.
- Quantity (0.65) and Unit Price (0.69) showed meaningful relationships with Total Sales.
- Statistical testing provides additional evidence for evaluating business assumptions.
- Data-driven pricing, promotions and category strategies can support better decisions

Data tells us what happened. Statistical validation helps us understand how confident we can be—and business strategy turns those insights into action